Three characterizations of an isomorphism of precategories

The Univalent Foundations Program

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Editorial note. The counted statement is the author's ct:isoprecat (source lines 693-703), whose third enumerated clause is labelled and whose displayed equation inside the statement carries the label eq:ct:isoprecattri; the proof environment follows one remark later at line 707, so the pairing is by adjacency around that remark.

Editorial note. The remark at line 705 is retained as context because it is the author's own qualification of the counted statement, and because it refers to the labelled equation inside the statement, which is retained with it.

Editorial note. The definition of an isomorphism of precategories is retained as context because the first clause of the statement uses that notion verbatim.

Editorial note. The retained counter values (chapter 9, section 4, theorem offset 7) reproduce the first-edition numbering: the retained definition prints as Definition 9.4.8, the counted lemma as Lemma 9.4.9, and the two labelled equations inside its statement and proof keep 9.4.10 and 9.4.11.

Editorial note. Three references leave the retained ranges, to a section of the foundations part, to the chapter on equivalences and to the adjointification lemma of the same section; all three are declared as non-load-bearing source pointers, and no part of the counted proof is replaced by them.

Editorial note. No retained line refers to a figure, bibliography entry, included file or non-ASCII character, so the package declares no asset and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: the three characterizations of an isomorphism of precategories.

Allowed standard prerequisites (editorial).

- Precategories, functors and the functor precategory B^A. A precategory has a type of objects and hom-sets with composition and identities; functors preserve composition and identities, and natural transformations between functors A -> B form the hom-sets of the functor precategory B^A.. The statement is about an isomorphism of precategories, i.e. a functor that is fully faithful with an equivalence on objects, and the proof works with identities of functors in B^A (source lines 740-750).

- idtoiso and isotoid in a precategory. For a precategory A and a, b : A the canonical map (a = b) -> (a iso b) is written idtoiso; in a category it is an equivalence with inverse isotoid, and it takes inverses to inverses and concatenation to composition.. The proof builds the action of the inverse functor on hom-sets by conjugating with idtoiso of the object-level isomorphism eps_0 (source lines 718-723) and uses its naturality at lines 757-763.
- Half-adjoint equivalences and the equivalence of the triangle identities. An equivalence of types may be presented by a half adjoint equivalence, whose triangle identity is exactly the object part of equation (9.4.10); the two triangle identities are equivalent, as shown in the chapter on equivalences.. Source lines 709-714 identify (9.4.10) with the half-adjoint triangle identity, and line 789 invokes the equivalence of the two triangle identities for the last round trip.

Source reference. Original section 2.4: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1609–1609. The retained context sentence refers to the earlier section where the convention for the word isomorphism is discussed. It is a navigation pointer only and no part of the counted statement or proof depends on it.

Source reference. Original section 4: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 164–164. The retained proof recalls that the two triangle identities are equivalent, a fact established in the chapter on equivalences. It is a navigation pointer to a standard prerequisite: the counted proof is retained line for line and no step of it is replaced by this reference.

Source reference. Original lemma 9.4.2: sources/hott-categories/pinned-original-source.tex, source lines 559–559. The last step of the retained proof turns a natural isomorphism into an equivalence of precategories through the adjointification lemma of the same section. It is a navigation pointer to a result available in the same source file: the counted lemma's own author proof is retained line for line and the pointer replaces no step of it.

Definition 9.4.8. A functor $F : A \to B$ is an **isomorphism of (pre)categories** if F is fully faithful and $F_0 : A_0 \to B_0$ is an equivalence of types.

This definition is an exception to our general rule (see original section 2.4) of only using the word "isomorphism" for sets and set-like objects. However, it does carry an appropriate connotation here, because for general precategories, isomorphism is stronger than equivalence.

Note that being an isomorphism of precategories is always a mere property. Let $A \cong B$ denote the type of isomorphisms of (pre)categories from A to B.

Lemma 9.4.9. *For precategories A and B and $F : A \to B$, the following are equivalent.*

(i) F is an isomorphism of precategories.
(ii) There exist $G : B \to A$ and $\eta : 1_A = GF$ and $\epsilon : FG = 1_B$ such that

$$\mathsf{ap}_{(\lambda H.\, FH)}(\eta) = \mathsf{ap}_{(\lambda K.\, KF)}(\epsilon^{-1}). \tag{9.4.10}$$

(iii) There merely exist $G : B \to A$ *and* $\eta : 1_A = GF$ *and* $\epsilon : FG = 1_B$.

Note that if B_0 is not a 1-type, then (9.4.10) may not be a mere proposition.

Proof. First note that since hom-sets are sets, equalities between equalities of functors are uniquely determined by their object-parts. Thus, by function extensionality, (9.4.10) is equivalent to

$$(F_0)(\eta_0)_a = (\epsilon_0)^{-1}{}_{F_0 a}. \tag{9.4.11}$$

for all $a : A_0$. Note that this is precisely the triangle identity for G_0, η_0, and ϵ_0 to be a proof that F_0 is a half adjoint equivalence of types.

Now suppose (i). Let $G_0 : B_0 \to A_0$ be the inverse of F_0, with $\eta_0 : \mathsf{id}_{A_0} = G_0 F_0$ and $\epsilon_0 : F_0 G_0 = \mathsf{id}_{B_0}$ satisfying the triangle identity, which is precisely (9.4.11). Now define $G_{b,b'} : \hom_B(b, b') \to \hom_A(G_0 b, G_0 b')$ by

$$G_{b,b'}(g) :\equiv (F_{G_0 b, G_0 b'})^{-1}\Big(\mathsf{idtoiso}((\epsilon_0)^{-1}{}_{b'}) \circ g \circ \mathsf{idtoiso}((\epsilon_0)_b)\Big)$$

(using the assumption that F is fully faithful). Since $\mathsf{idtoiso}$ takes inverses to inverses and concatenation to composition, and F is a functor, it follows that G is a functor.

By definition, we have $(GF)_0 \equiv G_0 F_0$, which is equal to id_{A_0} by η_0. To obtain $1_A = GF$, we need to show that when transported along η_0, the identity function of $\hom_A(a, a')$ becomes equal to the composite $G_{Fa,Fa'} \circ F_{a,a'}$. In other words, for any $f : \hom_A(a, a')$ we must have

$$\begin{aligned}&\mathsf{idtoiso}((\eta_0)_{a'}) \circ f \circ \mathsf{idtoiso}((\eta_0)^{-1}{}_a)\\ &\qquad = (F_{GFa,GFa'})^{-1}\Big(\mathsf{idtoiso}((\epsilon_0)^{-1}{}_{Fa'}) \circ F_{a,a'}(f) \circ \mathsf{idtoiso}((\epsilon_0)_{Fa})\Big).\end{aligned}$$

But this is equivalent to

$$\begin{aligned}&(F_{GFa,GFa'})\Big(\mathsf{idtoiso}((\eta_0)_{a'}) \circ f \circ \mathsf{idtoiso}((\eta_0)^{-1}{}_a)\Big)\\ &\qquad = \mathsf{idtoiso}((\epsilon_0)^{-1}{}_{Fa'}) \circ F_{a,a'}(f) \circ \mathsf{idtoiso}((\epsilon_0)_{Fa}).\end{aligned}$$

which follows from functoriality of F, the fact that F preserves $\mathsf{idtoiso}$, and (9.4.11). Thus we have $\eta : 1_A = GF$.

On the other side, we have $(FG)_0 \equiv F_0 G_0$, which is equal to id_{B_0} by ϵ_0. To obtain $FG = 1_B$, we need to show that when transported along ϵ_0, the identity function of $\hom_B(b, b')$ becomes equal to the composite $F_{Gb,Gb'} \circ G_{b,b'}$. That is, for any $g : \hom_B(b, b')$ we must have

$$\begin{aligned}&F_{Gb,Gb'}\Big((F_{Gb,Gb'})^{-1}\Big(\mathsf{idtoiso}((\epsilon_0)^{-1}{}_{b'}) \circ g \circ \mathsf{idtoiso}((\epsilon_0)_b)\Big)\Big)\\ &\qquad = \mathsf{idtoiso}((\epsilon_0{}^{-1})_{b'}) \circ g \circ \mathsf{idtoiso}((\epsilon_0)_b).\end{aligned}$$

But this is just the fact that $(F_{Gb,Gb'})^{-1}$ is the inverse of $F_{Gb,Gb'}$. And we have remarked that (9.4.10) is equivalent to (9.4.11), so (ii) holds.

Conversely, suppose given (ii); then the object-parts of G, η, and ϵ together with (9.4.11) show that F_0 is an equivalence of types. And for $a, a' : A_0$, we define $\overline{G}_{a,a'} : \hom_B(Fa, Fa') \to \hom_A(a, a')$ by

$$\overline{G}_{a,a'}(g) :\equiv \mathsf{idtoiso}(\eta^{-1})_{a'} \circ G(g) \circ \mathsf{idtoiso}(\eta)_a. \tag{9.4.12}$$

By naturality of $\mathsf{idtoiso}(\eta)$, for any $f : \hom_A(a, a')$ we have

$$\begin{aligned}\overline{G}_{a,a'}(F_{a,a'}(f)) &= \mathsf{idtoiso}(\eta^{-1})_{a'} \circ G(F(f)) \circ \mathsf{idtoiso}(\eta)_a \\ &= \mathsf{idtoiso}(\eta^{-1})_{a'} \circ \mathsf{idtoiso}(\eta)_{a'} \circ f \\ &= f.\end{aligned}$$

On the other hand, for $g : \hom_B(Fa, Fa')$ we have

$$\begin{aligned}F_{a,a'}(\overline{G}_{a,a'}(g)) &= F(\mathsf{idtoiso}(\eta^{-1})_{a'}) \circ F(G(g)) \circ F(\mathsf{idtoiso}(\eta)_a) \\ &= \mathsf{idtoiso}(\epsilon)_{Fa'} \circ F(G(g)) \circ \mathsf{idtoiso}(\epsilon^{-1})_{Fa} \\ &= \mathsf{idtoiso}(\epsilon)_{Fa'} \circ \mathsf{idtoiso}(\epsilon^{-1})_{Fa'} \circ g \\ &= g.\end{aligned}$$

(There are lemmas needed here regarding the compatibility of $\mathsf{idtoiso}$ and whiskering, which we leave it to the reader to state and prove.) Thus, $F_{a,a'}$ is an equivalence, so F is fully faithful; i.e. (i) holds.

Now the composite (i)→(ii)→(i) is equal to the identity since (i) is a mere proposition. On the other side, tracing through the above constructions we see that the composite (ii)→(i)→(ii) essentially preserves the object-parts G_0, η_0, ϵ_0, and the object-part of (9.4.10). And in the latter three cases, the object-part is all there is, since hom-sets are sets.

Thus, it suffices to show that we recover the action of G on hom-sets. In other words, we must show that if $g : \hom_B(b, b')$, then

$$G_{b,b'}(g) = \overline{G}_{G_0 b, G_0 b'}\Big(\mathsf{idtoiso}((\epsilon_0)^{-1}{}_{b'}) \circ g \circ \mathsf{idtoiso}((\epsilon_0)_b)\Big)$$

where $\overline{G}$ is defined by (9.4.12). However, this follows from functoriality of G and the *other* triangle identity, which we have seen in original section 4 is equivalent to (9.4.11).

Now since (i) is a mere proposition, so is (ii), so it suffices to show they are logically equivalent to (iii). Of course, (ii)→(iii), so let us assume (iii). Since (i) is a mere proposition, we may assume given G, η, and ϵ. Then G_0 along with η and ϵ imply that F_0 is an equivalence. Moreover, we also have natural isomorphisms $\mathsf{idtoiso}(\eta) : 1_A \cong GF$ and $\mathsf{idtoiso}(\epsilon) : FG \cong 1_B$, so by original lemma 9.4.2, F is an equivalence of precategories, and in particular fully faithful. □